

mance that is possible with the colored measurement-noise filters described in this section.

Table 7.2 Experimental values of the trace of the covariance of the estimation error (500 time steps) for Example 7.2. As the color content of the measurement noise increases (i.e., as ψ increases) the colored measurement-noise filters provide increasingly better performance than the standard Kalman filter

Color ψ	Standard Filter	Augmented Filter	Measurement Differencing
0.0	0.245	0.245	0.247
0.2	0.260	0.258	0.259
0.5	0.308	0.294	0.295
0.9	0.631	0.407	0.406

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7.3 STEADY-STATE FILTERING

Many Kalman filter implementations are coded in embedded systems (rather than desktop computers) in which memory and computational effort is still a primary consideration. If the underlying system is time-invariant, and the process- and measurement-noise covariances are time-invariant, then we can replace the time-varying Kalman filter of Chapter 5 with a steady-state Kalman filter. The steady-state filter often performs nearly as well as the time-varying filter. Using a steady-state filter has the advantage that we do not have to compute the estimation-error covariance or Kalman gain in real time. Note that a steady-state Kalman filter is still a dynamic system. The term “steady-state” Kalman filtering means that the Kalman filter is time-invariant; it is the Kalman gain that is in steady state.

As an example, recall the scalar system discussed in Example 5.2:

$$\begin{aligned}
 x_{k+1} &= x_k + w_k \\
 y_k &= x_k + v_k \\
 w_k &\sim (0, 1) \\
 v_k &\sim (0, 1)
 \end{aligned} \tag{7.36}$$

We saw from Example 5.2 that the Kalman gain converged to a steady-state value after a few time steps:

$$\begin{aligned}
 \lim_{k \rightarrow \infty} K_k &= K_\infty \\
 &= \frac{1 + \sqrt{5}}{3 + \sqrt{5}}
 \end{aligned} \tag{7.37}$$

So instead of performing the measurement-update equation for P_k , the time-update equation for P_k , and the Kalman gain computation for K_k at each time step, we

can simply use the constant K_∞ as our Kalman gain at each time step. For a system with many states, this can save a lot of computational effort, especially considering the fact that this will allow us to avoid real-time matrix inversions. The steady-state Kalman filter for this example is simply given as

$$\begin{aligned}\hat{x}_k^- &= F\hat{x}_{k-1}^+ \\ \hat{x}_k^+ &= \hat{x}_k^- + K_\infty(y_k - H\hat{x}_k^-) \\ &= F\hat{x}_{k-1}^+ + K_\infty(y_k - HF\hat{x}_{k-1}^+) \\ &= (I - K_\infty H)F\hat{x}_{k-1}^+ + K_\infty y_k\end{aligned}\quad (7.38)$$

The steady-state Kalman filter is not optimal because we are not using the optimal Kalman gain at each time step (although it approaches optimality in the limit as $k \rightarrow \infty$). We are instead using the steady-state Kalman gain. However, for many problems of practical interest, the performance of the steady-state filter is nearly indistinguishable from that of the time-varying filter. For any particular problem, the difference between the time-varying and steady-state filters needs to be assessed by simulation or experimental results.

One way to determine the steady-state Kalman gain is by numerical simulation. We can simply write a computer program to propagate the Kalman gain as a function of time, and then observe the value toward which the gain is converging.

Another way to determine the steady-state Kalman gain is to manipulate the Kalman filter equations from Equation (7.14). Recall the covariance time-update equation for a time-invariant system:

$$P_{k+1}^- = FP_k^+ F^T + Q \quad (7.39)$$

Now substitute the expression for P_k^+ from Equation (7.14) into this equation to obtain

$$P_{k+1}^- = FP_k^- F^T - FK_k HP_k^- F^T - FK_k M^T F^T + Q \quad (7.40)$$

Now substitute the expression for K_k from Equation (7.14) into this equation to obtain

$$\begin{aligned}P_{k+1}^- &= FP_k^- F^T - \\ &\quad F(P_k^- H^T + M)(HP_k^- H^T + HM + M^T H^T + R)^{-1} HP_k^- F^T - \\ &\quad F(P_k^- H^T + M)(HP_k^- H^T + HM + M^T H^T + R)^{-1} M^T F^T + Q \\ &= FP_k^- F^T - F(P_k^- H^T + M)(HP_k^- H^T + HM + M^T H^T + R)^{-1} \times \\ &\quad (HP_k^- + M^T) F^T + Q\end{aligned}\quad (7.41)$$

If P_k^- converges to a steady-state value, then $P_k^- = P_{k+1}^-$ for large k . We will denote this steady-state value as P_∞ , which means that we can write

$$\begin{aligned}P_\infty &= FP_\infty F^T - \\ &\quad F(P_\infty H^T + M)(HP_\infty H^T + HM + M^T H^T + R)^{-1} \times \\ &\quad (HP_\infty + M^T) F^T + Q\end{aligned}\quad (7.42)$$

This is called an algebraic Riccati equation (ARE), or more specifically a discrete ARE (DARE).¹ Once we have P_∞ , we can substitute it for P_k^- in the Kalman gain

¹In MATLAB's Control System Toolbox, we can solve this equation by invoking the command `DARE(FT, HT, Q, HM + MTHT + R, FM)`.

formula of Equation (7.14) to obtain the steady-state Kalman gain:

$$K_{\infty} = (P_{\infty}H^T + M)(HP_{\infty}H^T + HM + M^TH^T + R)^{-1} \quad (7.43)$$

There are systems for which the Riccati equation (and hence the Kalman gain) does *not* converge to a steady-state value. Furthermore, it may converge to different steady-state values depending on the initial condition P_0 . Finally, even when it does converge to a steady-state value, it may result in an unstable Kalman filter. These issues comprise a rich field of study that has been reported widely in many books and papers [McG74, And79, Kai81, Goo84, Chu87]. We will summarize the most important Riccati equation convergence results below, but first we need to define what it means for a system to be controllable on the unit circle.

Definition 11 *The matrix pair (F, G) is controllable on the unit circle if there exists some matrix K such that $(F - GK)$ does not have any eigenvalues with magnitude 1.*

We illustrate this definition with some simple examples.

■ EXAMPLE 7.3

Consider the scalar system

$$x_{k+1} = x_k \quad (7.44)$$

In this example, $F = 1$ and $G = 0$. The system dynamics are independent of any control signal, and the system has an eigenvalue with a magnitude of 1. The system is not controllable on the unit circle because its eigenvalue has a magnitude of 1 regardless of the feedback control input.

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■ EXAMPLE 7.4

Consider the scalar system

$$x_{k+1} = 2x_k \quad (7.45)$$

In this example, $F = 2$ and $G = 0$. As in the previous example, the system dynamics are independent of any control signal. However, the system eigenvalue has a magnitude of 2. The system is controllable on the unit circle because there exists a feedback control gain K such that $(F - GK)$ does not have any eigenvalues with a magnitude of 1. In fact, regardless of the feedback control gain, the system eigenvalues will never have a magnitude of 1.

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■ EXAMPLE 7.5

Consider the system

$$x_{k+1} = \begin{bmatrix} F_1 & 0 \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u_k \quad (7.46)$$

When the feedback control $u_k = -Kx_k$ is implemented, where $K = \begin{bmatrix} K_1 & K_2 \end{bmatrix}$, the closed-loop system becomes

$$x_{k+1} = \begin{bmatrix} F_1 & 0 \\ -K_1 & 1 - K_2 \end{bmatrix} x_k \quad (7.47)$$

The closed-loop system has eigenvalues at F_1 and $(1 - K_2)$. We see that if $F_1 = \pm 1$ then there is no feedback control gain K that results in all closed-loop eigenvalues having a nonunity magnitude, and the system is therefore not controllable on the unit circle. However, if $F_1 \neq \pm 1$, then we can find a feedback control gain K that does result in all closed-loop eigenvalues having a nonunity magnitude, and the system is therefore controllable on the unit circle.

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Next we summarize the most important Riccati equation convergence results from [Bit85, Pou86, Kai00], where proofs are given. Recall that the DARE is given as

$$\begin{aligned} P_\infty = & FP_\infty F^T - \\ & F(P_\infty H^T + M)(HP_\infty H^T + HM + M^T H^T + R)^{-1} \times \\ & (HP_\infty + M^T)F^T + Q \end{aligned} \quad (7.48)$$

We assume that $Q \geq 0$ and $R > 0$. We define G as any matrix such that $GG^T = Q - MR^{-1}M^T$. The corresponding steady-state Kalman gain K_∞ is given as

$$K_\infty = (P_\infty H^T + M)(HP_\infty H^T + HM + M^T H^T + R)^{-1} \quad (7.49)$$

The steady-state Kalman filter is given as

$$\hat{x}_k^+ = (I - K_\infty H)F\hat{x}_{k-1}^+ + K_\infty y_k \quad (7.50)$$

We say that the DARE solution P_∞ is stabilizing if it results in a stable steady-state filter. That is, P_∞ is defined as a stabilizing DARE solution if all of the eigenvalues of $(I - K_\infty H)F$ are less than one in magnitude.

Theorem 23 *The DARE has a unique positive semidefinite solution P_∞ if and only if both of the following conditions hold.*

1. (F, H) is detectable.
2. $(F - MR^{-1}H, G)$ is stabilizable.

Furthermore, the corresponding steady-state Kalman filter is stable. That is, the eigenvalues of $(I - K_\infty H)F$ have magnitude less than 1.

Theorem 23 does not preclude the existence of DARE solutions that are negative definite or indefinite. If such solutions exist, then they would result in an unstable Kalman filter. If we weaken the stabilizability condition in Theorem 23, we obtain the following.

Theorem 24 *The DARE has at least one positive semidefinite solution P_∞ if and only if both of the following conditions hold.*

1. (F, H) is detectable.
2. $(F - MR^{-1}H, G)$ is controllable on the unit circle.

Furthermore, exactly one of the positive semidefinite DARE solutions results in a stable steady-state Kalman filter.

Since controllability on the unit circle is a subset of stabilizability, we see that Theorem 24 is a subset of Theorem 23. Theorem 24 states conditions for the existence of exactly one stabilizing positive semidefinite DARE solution. However, there may be additional DARE solutions (positive semidefinite or otherwise) that result in unstable Kalman filters. If a time-varying Kalman filter is run in this situation, then the Kalman filter equations may converge to either a stable or an unstable filter, depending on the initial condition P_0^+ . If we strengthen the controllability condition of Theorem 24, we obtain the following.

Theorem 25 *The DARE has at least one positive definite solution P_∞ if and only if both of the following conditions hold.*

1. (F, H) is detectable.
2. $(F - MR^{-1}H, G)$ is controllable on and inside the unit circle.

Furthermore, exactly one of the positive definite DARE solutions results in a stable steady-state Kalman filter.

If we drop the controllability condition in the above two theorems, we obtain the following.

Theorem 26 *The DARE has at least one positive semidefinite solution P_∞ if (F, H) is detectable. Furthermore, at least one such solution results in a marginally stable steady-state Kalman filter.*

Note that the resulting filter is only marginally stable, so it may have eigenvalues on the unit circle. Also note that this theorem poses a sufficient (not necessary) condition. That is, there may be a stable steady-state Kalman filter even if the conditions of the above theorem do not hold. Furthermore, even if the conditions of the theorem do hold, there may be DARE solutions that result in unstable Kalman filters.

■ EXAMPLE 7.6

Consider again the scalar system of Equation (7.36). We see that $F = 1$, $H = 1$, $Q = 1$, $R = 1$, and $M = 0$. Note that (F, H) is observable, and (F, G) is controllable for all G such that $GG^T = Q$ (recall that $M = 0$ for this example). We therefore know from Theorem 23 that the DARE has a unique positive semidefinite solution. We know from Theorem 25 that the DARE solution is not only positive semidefinite, but it is also positive definite. We also know from these two theorems that the corresponding steady-state Kalman filter is stable. The DARE for this system is given by

$$\begin{aligned} P &= FPF^T - FPH^T(HPH^T + R)^{-1}HPF^T + Q \\ &= P - P(P + 1)^{-1}P + 1 \end{aligned} \tag{7.51}$$

This can be solved to obtain

$$P = \frac{1 \pm \sqrt{5}}{2} \quad (7.52)$$

So the DARE has two solutions, one of which is negative and one of which is positive. If we use the negative DARE solution in the steady-state Kalman filter we obtain

$$\begin{aligned} K &= PH^T(HPH^T + R)^{-1} \\ &= \frac{1 - \sqrt{5}}{3 - \sqrt{5}} \\ \hat{x}_k^+ &= (I - KH)F\hat{x}_{k-1}^+ + Ky_k \\ &= \frac{2}{3 - \sqrt{5}}\hat{x}_{k-1}^+ + Ky_k \\ &\approx 2.62\hat{x}_{k-1}^+ + Ky_k \end{aligned} \quad (7.53)$$

We see that the resulting Kalman filter is unstable. However, if we use the positive DARE solution in the steady-state Kalman filter we obtain

$$\begin{aligned} K &= \frac{1 + \sqrt{5}}{3 + \sqrt{5}} \\ \hat{x}_k^+ &= \frac{2}{3 + \sqrt{5}}\hat{x}_{k-1}^+ + Ky_k \\ &\approx 0.38\hat{x}_{k-1}^+ + Ky_k \end{aligned} \quad (7.54)$$

We see that the resulting Kalman filter is stable.

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■ EXAMPLE 7.7

Consider a scalar system with $F = 1$, $H = 1$, $Q = 0$, $R = 1$, and $M = 0$. Note (F, H) is detectable. However, it is not true that (F, G) is controllable on the unit circle for all G such that $GG^T = Q$. We therefore know from Theorem 24 that the DARE does not have a positive semidefinite solution that results in a stable Kalman filter. However, we know from Theorem 26 that the DARE has a positive semidefinite solution that results in a marginally stable Kalman filter. The DARE for this system is given by

$$\begin{aligned} P &= FPF^T - FPH^T(HPH^T + R)^{-1}HPF^T + Q \\ &= P - P(P + 1)^{-1}P \end{aligned} \quad (7.55)$$

This has two solutions for P , both of which are 0 (i.e., positive semidefinite). If we use this solution in the steady-state Kalman filter we obtain

$$\begin{aligned} K &= 0 \\ \hat{x}_k^+ &= \hat{x}_{k-1}^+ \end{aligned} \quad (7.56)$$

We see that the resulting Kalman filter is marginally stable (the eigenvalue is 1).

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■ EXAMPLE 7.8

Consider a scalar system with $F = 2$, $H = 1$, $Q = 0$, $R = 1$, and $M = 0$. Note (F, H) is detectable. Also (F, G) is controllable on and inside the unit circle for all G such that $GG^T = Q$. We therefore know from Theorem 24 that the DARE has exactly one positive semidefinite solution that results in a stable Kalman filter.

However, we know from Theorem 26 that the DARE has exactly one positive semidefinite solution that results in a marginally stable Kalman filter is stable. We also know from Theorem 25 that this DARE solution is positive definite. The DARE for this system is given by

$$\begin{aligned} P &= FPF^T - FPH^T(HPH^T + R)^{-1}HPF^T + Q \\ &= 4P - 4P(P+1)^{-1}P \end{aligned} \quad (7.57)$$

This has two solutions for P , one of which is 0 (i.e., positive semidefinite), and one of which is 3 (i.e., positive definite). If we use $P = 0$ in the steady-state Kalman filter we obtain

$$\begin{aligned} K &= 0 \\ \hat{x}_k^+ &= 2\hat{x}_{k-1}^+ \end{aligned} \quad (7.58)$$

We see that the resulting Kalman filter is unstable (the eigenvalue is 2). If we use $P = 3$ in the steady-state Kalman filter we obtain

$$\begin{aligned} K &= \frac{3}{4} \\ \hat{x}_k^+ &= \frac{1}{2}\hat{x}_{k-1}^+ \end{aligned} \quad (7.59)$$

We see that the resulting Kalman filter is stable (the eigenvalue is 1/2). In this example, we have multiple positive semidefinite solutions to the DARE, but only one results in a stable Kalman filter.

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7.3.1 α - β filtering

In this section, we derive the α - β filter [Bar01], also sometimes referred to as the f - g filter or the g - h filter [Bro98]. The α - β filter is a steady-state Kalman filter that is applied to a two-state Newtonian system with a position measurement. This is the type of estimation problem that commonly arises in tracking problems, and so it is well known and has been widely studied since before the invention of the Kalman filter.

Suppose we have a Newtonian dynamic system with only two states (position and velocity) and a noisy acceleration input, and we measure position plus noise. The system and measurement equations are then given as

$$\begin{aligned} x_k &= \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} x_{k-1} + \begin{bmatrix} T^2/2 \\ T \end{bmatrix} w'_{k-1} \\ y_k &= \begin{bmatrix} 1 & 0 \end{bmatrix} x_k + v_k \end{aligned}$$

$$\begin{aligned} w'_k &\sim (0, \sigma_w^2) \\ v_k &\sim (0, R) \end{aligned} \quad (7.60)$$

where T is the sample time, and w'_k and v_k are uncorrelated white noise processes. The process equation can be written as

$$\begin{aligned} x_k &= \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} x_{k-1} + w_{k-1} \\ w_k &\sim (0, Q) \\ Q &= \begin{bmatrix} T^2/2 \\ T \end{bmatrix} E[w'_k w_k'^T] \begin{bmatrix} T^2/2 & T \end{bmatrix} \\ &= \begin{bmatrix} T^4/4 & T^3/2 \\ T^3/2 & T^2 \end{bmatrix} \sigma_w^2 \end{aligned} \quad (7.61)$$

A steady-state Kalman filter can be designed for this system from Equation (5.19), which is repeated here using steady-state notation:

$$\begin{aligned} P^- &= FP^+F^T + Q \\ K &= P^-H^T(HP^-H^T + R)^{-1} \\ \hat{x}_k^- &= F\hat{x}_{k-1}^+ \\ \hat{x}_k^+ &= \hat{x}_k^- + K(y_k - H_k\hat{x}_k^-) \\ P^+ &= (I - KH)P^- \end{aligned} \quad (7.62)$$

For this two-state, one-measurement problem, we see that K is a 2×1 matrix, and P^- and P^+ are 2×2 matrices. We will denote their steady-state values as

$$\begin{aligned} K &= \begin{bmatrix} K_1 & K_2 \end{bmatrix}^T \\ &= \begin{bmatrix} \alpha & \beta/T \end{bmatrix}^T \\ P^- &= \begin{bmatrix} P_{11}^- & P_{12}^- \\ P_{12}^- & P_{22}^- \end{bmatrix} \\ P^+ &= \begin{bmatrix} P_{11}^+ & P_{12}^+ \\ P_{12}^+ & P_{22}^+ \end{bmatrix} \end{aligned} \quad (7.63)$$

The parameters of the Kalman gain matrix K define the α and β parameters of the α - β filter. We can use Equation (7.62) to write

$$\begin{aligned} K &= \begin{bmatrix} P_{11}^- & P_{12}^- \\ P_{12}^- & P_{22}^- \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} P_{11}^- & P_{12}^- \\ P_{12}^- & P_{22}^- \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + R \right)^{-1} \\ &= \frac{1}{P_{11}^- + R} \begin{bmatrix} P_{11}^- & P_{12}^- \end{bmatrix}^T \end{aligned} \quad (7.64)$$

The P^+ expression in Equation (7.62) can be written as

$$\begin{aligned} \begin{bmatrix} P_{11}^+ & P_{12}^+ \\ P_{12}^+ & P_{22}^+ \end{bmatrix} &= \left(I - \begin{bmatrix} K_1 \\ K_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \right) \begin{bmatrix} P_{11}^- & P_{12}^- \\ P_{12}^- & P_{22}^- \end{bmatrix} \\ &= \begin{bmatrix} (1 - K_1)P_{11}^- & (1 - K_1)P_{12}^- \\ (1 - K_1)P_{12}^- & P_{22}^- - K_2P_{12}^- \end{bmatrix} \end{aligned} \quad (7.65)$$

The P^- expression in Equation (7.62) can be rewritten in terms of P^+ as follows:

$$\begin{aligned}
 P^+ &= F^{-1}(P^- - Q)F^{-T} \\
 \begin{bmatrix} P_{11}^+ & P_{12}^+ \\ P_{12}^+ & P_{22}^+ \end{bmatrix} &= \begin{bmatrix} 1 & -T \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} P_{11}^- & P_{12}^- \\ P_{12}^- & P_{22}^- \end{bmatrix} - \begin{bmatrix} T^4/4 & T^3/2 \\ T^3/2 & T^2 \end{bmatrix} \sigma_w^2 \right) \times \\
 &\quad \begin{bmatrix} 1 & 0 \\ -T & 1 \end{bmatrix} \tag{7.66}
 \end{aligned}$$

Carrying out the multiplication gives the elements of P^+ as

$$\begin{aligned}
 P_{12}^+ &= P_{12}^- + \sigma_w^2 T^3/2 - P_{22}^- T \\
 P_{11}^+ &= P_{11}^- + \sigma_w^2 T^4/4 - P_{12}^- T - P_{12}^+ T \\
 P_{22}^+ &= P_{22}^- - \sigma_w^2 T^2 \tag{7.67}
 \end{aligned}$$

Equating the P_{ij}^+ elements in Equations (7.65) and (7.67) and performing a little algebra gives

$$\begin{aligned}
 K_1 P_{11}^- &= 2T P_{12}^- - T^2 P_{22}^- + T^4 \sigma_w^2/4 \\
 K_1 P_{12}^- &= T P_{22}^- - T^3 \sigma_w^2/2 \\
 K_2 P_{12}^- &= T^2 \sigma_w^2 \tag{7.68}
 \end{aligned}$$

These three equations, along with the expressions for K_1 and K_2 in the last line of Equation (7.64), can be solved for the five unknowns K_1 , K_2 , P_{11}^- , P_{12}^- , and P_{22}^- . After some algebra, this gives

$$\begin{aligned}
 K_1 &= -\frac{1}{8} \left(\lambda^2 + 8\lambda - (\lambda + 4)\sqrt{\lambda^2 + 8\lambda} \right) \\
 K_2 &= \frac{1}{4T} \left(\lambda^2 + 4\lambda - \lambda\sqrt{\lambda^2 + 8\lambda} \right) \\
 P_{11}^- &= \frac{K_1 \sigma_w^2}{1 - K_1} \\
 P_{12}^- &= \frac{K_2 \sigma_w^2}{1 - K_1} \\
 P_{22}^- &= \left(\frac{K_1}{T} + \frac{K_2}{2} \right) P_{12}^- \tag{7.69}
 \end{aligned}$$

where λ is called the target maneuvering index or target tracking index [Kal84] and is defined as

$$\lambda = \frac{\sigma_w^2 T^2}{R} \tag{7.70}$$

Note that λ gives the ratio of the motion uncertainty to the measurement uncertainty. From these expressions and Equation (7.65) it can be shown that the elements of the steady-state *a posteriori* estimation-error covariance are given as

$$\begin{aligned}
 P_{11}^+ &= K_1 R \\
 P_{12}^+ &= K_2 R \\
 P_{22}^+ &= \left(\frac{K_1}{T} - \frac{K_2}{2} \right) P_{12}^- \tag{7.71}
 \end{aligned}$$

7.3.2 α - β - γ filtering

In this section, we present (without derivation) the α - β - γ filter [Bar01], also sometimes referred to as the f - g - h filter or the g - h - k filter [Bro98]. The α - β - γ filter is a steady-state Kalman filter that is applied to a three-state Newtonian system with a position measurement. This is very similar to the α - β filter presented in the previous section, except that the dynamic system model is one order higher in the α - β - γ filter.

Consider the three-state system given in Example 5.1. The states consist of position, velocity, and acceleration, the input consists of noisy acceleration, and the measurement consists of position plus noise. The system and measurement equations are given as

$$\begin{aligned} x_k &= \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} x_{k-1} + \begin{bmatrix} T^2/2 \\ T \\ 1 \end{bmatrix} w'_{k-1} \\ y_k &= [1 \ 0 \ 0] x_k + v_k \\ w'_k &\sim (0, \sigma_w^2) \\ v_k &\sim (0, R) \end{aligned} \quad (7.72)$$

where T is the sample time, and w'_k and v_k are uncorrelated white noise processes. The process equation can be written as

$$\begin{aligned} x_k &= \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} x_{k-1} + w_{k-1} \\ w_k &\sim (0, Q) \\ Q &= \begin{bmatrix} T^2/2 \\ T \\ 1 \end{bmatrix} E[w'_k w_k^T] \begin{bmatrix} T^2/2 & T & 1 \end{bmatrix} \\ &= \begin{bmatrix} T^4/4 & T^3/2 & T^2/2 \\ T^3/2 & T^2 & T \\ T^2/2 & T & 1 \end{bmatrix} \sigma_w^2 \end{aligned} \quad (7.73)$$

A steady-state Kalman filter can be designed for this system from Equation (5.19), in a similar way that the α - β filter was designed in the previous section. The steady-state values of the Kalman gain and *a posteriori* estimation-error covariance are denoted as

$$\begin{aligned} K &= [K_1 \ K_2 \ K_3]^T \\ &= [\alpha \ \beta/T \ \gamma/2T^2]^T \\ P^+ &= \begin{bmatrix} P_{11}^+ & P_{12}^+ & P_{13}^+ \\ P_{12}^+ & P_{22}^+ & P_{23}^+ \\ P_{13}^+ & P_{23}^+ & P_{33}^+ \end{bmatrix} \end{aligned} \quad (7.74)$$

The parameters of the Kalman gain matrix K define the α , β , and γ parameters of the α - β - γ filter. The solution can be computed as follows [Gra93]:

$$\begin{aligned}
\alpha &= 1 - s^2 \\
\beta &= 2(1 - s)^2 \\
\gamma &= 2\lambda s
\end{aligned} \tag{7.75}$$

where λ is the target maneuvering index defined in Equation (7.70), and s is an auxiliary variable. The variable s is defined via auxiliary variables b , c , p , q , and z as follows.

$$\begin{aligned}
b &= \frac{\lambda}{2} - 3 \\
c &= \frac{\lambda}{2} + 3 \\
p &= c - \frac{b^2}{3} \\
q &= \frac{2b^3}{27} - \frac{bc}{3} - 1 \\
z &= \left[\frac{-q + \sqrt{q^2 + 4p^3/27}}{2} \right]^{1/3} \\
s &= z - \frac{p}{3z} - \frac{b}{3}
\end{aligned} \tag{7.76}$$

The steady-state *a posteriori* error covariance can be computed as

$$\begin{aligned}
P_{11}^+ &= \alpha R \\
P_{12}^+ &= \beta R/T \\
P_{13}^+ &= \gamma R/2T^2 \\
P_{22}^+ &= \frac{8\alpha\beta + \gamma(\beta - 2\alpha - 4)}{8T^2(1 - \alpha)} R \\
P_{23}^+ &= \frac{\beta(2\beta - \gamma)R}{4T^3(1 - \alpha)} \\
P_{33}^+ &= \frac{\gamma(2\beta - \gamma)R}{4T^4(1 - \alpha)}
\end{aligned} \tag{7.77}$$

The general idea of the α - β and α - β - γ filters date back to the 1940s [Mec49, Skl57, Ben62], before the advent of Kalman filtering, although, of course, the optimal α - β - γ values were not known at that time. Further discussion of these filters and related issues can be found in [Bro98, Bar01]. A steady-state Kalman filter that is applied to a one-state Newtonian system with a position measurement is called an α filter [Sio96].

7.3.3 A Hamiltonian approach to steady-state filtering

In this section, we present an alternative method for obtaining the steady-state Kalman filter. We will assume in this section that the correlation M between the process noise and measurement noise is zero so that we can simplify notation. The

a priori Riccati equation of Equation (7.41) can then be written as

$$P_{k+1} = F P_k F^T - F P_k H^T (H P_k H^T + R)^{-1} H P_k F^T + Q \quad (7.78)$$

where we have dropped the minus superscript for ease of notation. We can use the matrix inversion lemma of Equation (1.39) to write

$$(H P_k H^T + R)^{-1} = R^{-1} - R^{-1} H (H^T R^{-1} H + P_k^{-1})^{-1} H^T R^{-1} \quad (7.79)$$

Substituting this into Equation (7.78) gives

$$\begin{aligned} P_{k+1} = & F P_k F^T - F P_k H^T R^{-1} H P_k F^T + \\ & F P_k H^T R^{-1} H (H^T R^{-1} H + P_k^{-1})^{-1} H^T R^{-1} H P_k F^T + Q \end{aligned} \quad (7.80)$$

Factoring out F and F^T from the beginning and end of the first three terms on the right side gives

$$\begin{aligned} P_{k+1} = & F \{ P_k - P_k H^T R^{-1} H P_k + \\ & P_k H^T R^{-1} H (H^T R^{-1} H + P_k^{-1})^{-1} H^T R^{-1} H P_k \} F^T + Q \\ = & F \{ P_k - P_k H^T R^{-1} H [P_k - (H^T R^{-1} H + P_k^{-1})^{-1} H^T R^{-1} H P_k] \} F^T + Q \\ = & F \{ P_k - P_k H^T R^{-1} H (H^T R^{-1} H + P_k^{-1})^{-1} \times \\ & [(H^T R^{-1} H + P_k^{-1}) P_k - H^T R^{-1} H P_k] \} F^T + Q \\ = & F \{ P_k - P_k H^T R^{-1} H (H^T R^{-1} H + P_k^{-1})^{-1} \} F^T + Q \\ = & F P_k [(H^T R^{-1} H + P_k^{-1}) - H^T R^{-1} H] (H^T R^{-1} H + P_k^{-1})^{-1} F^T + Q \\ = & F (H^T R^{-1} H + P_k^{-1})^{-1} F^T + Q \\ = & F P_k (H^T R^{-1} H P_k + I)^{-1} F^T + Q F^{-T} F^T \\ = & [F P_k + Q F^{-T} (H^T R^{-1} H P_k + I)] (H^T R^{-1} H P_k + I)^{-1} F^T \\ = & [(F + Q F^{-T} H^T R^{-1} H) P_k + Q F^{-T}] (H^T R^{-1} H P_k + I)^{-1} F^T \end{aligned} \quad (7.81)$$

Now suppose that P_k can be factored as

$$P_k = S_k Z_k^{-1} \quad (7.82)$$

where S_k and Z_k both have the same dimensions as P_k . Making this substitution in Equation (7.81) gives

$$\begin{aligned} P_{k+1} = & [(F + Q F^{-T} H^T R^{-1} H) S_k + Q F^{-T} Z_k] Z_k^{-1} (H^T R^{-1} H S_k Z_k^{-1} + I)^{-1} F^T \\ = & [(F + Q F^{-T} H^T R^{-1} H) S_k + Q F^{-T} Z_k] (H^T R^{-1} H S_k + Z_k)^{-1} F^T \\ = & [(F + Q F^{-T} H^T R^{-1} H) S_k + Q F^{-T} Z_k] (F^{-T} H^T R^{-1} H S_k + F^{-T} Z_k)^{-1} \\ = & S_{k+1} Z_{k+1}^{-1} \end{aligned} \quad (7.83)$$

This shows that

$$\begin{aligned} S_{k+1} &= (F + Q F^{-T} H^T R^{-1} H) S_k + Q F^{-T} Z_k \\ Z_{k+1} &= F^{-T} H^T R^{-1} H S_k + F^{-T} Z_k \end{aligned} \quad (7.84)$$

These equations for S_{k+1} and Z_{k+1} can be written as the following single equation:

$$\begin{aligned} \begin{bmatrix} Z_{k+1} \\ S_{k+1} \end{bmatrix} &= \begin{bmatrix} F^{-T} & F^{-T}H^TR^{-1}H \\ QF^{-T} & F + QF^{-T}H^TR^{-1}H \end{bmatrix} \begin{bmatrix} Z_k \\ S_k \end{bmatrix} \\ &= \mathcal{H} \begin{bmatrix} Z_k \\ S_k \end{bmatrix} \end{aligned} \quad (7.85)$$

If the covariance matrix P is an $n \times n$ matrix, then $\mathcal{H}$ will be a $2n \times 2n$ matrix. The matrix $\mathcal{H}$ on the right side of the above equation is called a Hamiltonian matrix and has some interesting properties. It is a symplectic matrix; that is, it satisfies the equation

$$J^{-1}\mathcal{H}^T J = \mathcal{H}^{-1} \quad \text{where } J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \quad (7.86)$$

Symplectic matrices have the following properties (see Problem 7.7).

- None of the eigenvalues of a symplectic matrix are equal to 0.
- If λ is an eigenvalue of a symplectic matrix, then so is $1/\lambda$.
- The determinant of a symplectic matrix is equal to ± 1 .

If a symplectic matrix does not have any eigenvalues with magnitude equal to one, then half of its eigenvalues will be outside the unit circle, and the other half will be inside the unit circle. Let us define Λ as the diagonal matrix that contains all of the eigenvalues of $\mathcal{H}$ that are outside the unit circle (assuming that none of the eigenvalues are on the unit circle). Then the Jordan form of $\mathcal{H}$ can be written as

$$\begin{aligned} \mathcal{H} &= \Psi \begin{bmatrix} \Lambda^{-1} & 0 \\ 0 & \Lambda \end{bmatrix} \Psi^{-1} \\ &= \Psi D \Psi^{-1} \end{aligned} \quad (7.87)$$

where the D matrix is the diagonal matrix of eigenvalues, and is defined by the above equation. The Ψ matrix can be partitioned into four $n \times n$ blocks as

$$\Psi = \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix} \quad (7.88)$$

Note that the $2n \times n$ matrix $\begin{bmatrix} \Psi_{11} \\ \Psi_{21} \end{bmatrix}$ contains the eigenvectors of $\mathcal{H}$ that correspond to the stable eigenvalues of $\mathcal{H}$ (i.e., the eigenvalues that are inside the unit circle). The $2n \times n$ matrix $\begin{bmatrix} \Psi_{12} \\ \Psi_{22} \end{bmatrix}$ contain the eigenvectors of $\mathcal{H}$ that correspond to the unstable eigenvalues of $\mathcal{H}$ (i.e., the eigenvalues that are outside the unit circle). Equation (7.85) can be written as

$$\begin{aligned} \begin{bmatrix} Z_{k+1} \\ S_{k+1} \end{bmatrix} &= \Psi D \Psi^{-1} \begin{bmatrix} Z_k \\ S_k \end{bmatrix} \\ \Psi^{-1} \begin{bmatrix} Z_{k+1} \\ S_{k+1} \end{bmatrix} &= D \Psi^{-1} \begin{bmatrix} Z_k \\ S_k \end{bmatrix} \end{aligned} \quad (7.89)$$

Now define the $n \times n$ matrices Y_{1k} and Y_{2k} , and the $2n \times n$ matrix Y_k , as follows:

$$\begin{aligned} \begin{bmatrix} Y_{1k} \\ Y_{2k} \end{bmatrix} &= \Psi^{-1} \begin{bmatrix} Z_k \\ S_k \end{bmatrix} \\ &= \begin{bmatrix} (\Psi^{-1})_{11} & (\Psi^{-1})_{12} \\ (\Psi^{-1})_{21} & (\Psi^{-1})_{22} \end{bmatrix} \begin{bmatrix} Z_k \\ S_k \end{bmatrix} \\ &= Y_k \end{aligned} \quad (7.90)$$

Note in the above equation that $(\Psi^{-1})_{11}$ is *not* the inverse of the upper left $n \times n$ partition of Ψ ; the matrix $(\Psi^{-1})_{11}$ is rather the upper left $n \times n$ partition of Ψ^{-1} . (Similar statements hold for the other partitions.) With these definitions we can write Equation (7.89) as

$$\begin{aligned} Y_{k+1} &= DY_k \\ \begin{bmatrix} Y_{1,k+1} \\ Y_{2,k+1} \end{bmatrix} &= \begin{bmatrix} \Lambda^{-1} & 0 \\ 0 & \Lambda \end{bmatrix} \begin{bmatrix} Y_{1k} \\ Y_{2k} \end{bmatrix} \end{aligned} \quad (7.91)$$

From this equation we see that

$$\begin{aligned} Y_{2,k+1} &= \Lambda Y_{2k} \\ Y_{2k} &= \Lambda^k Y_{2,0} \end{aligned} \quad (7.92)$$

Similarly we see that

$$Y_{1k} = \Lambda^{-k} Y_{1,0} \quad (7.93)$$

Now note that Equation (7.90) can be written as

$$\begin{aligned} \begin{bmatrix} Z_k \\ S_k \end{bmatrix} &= \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix} \begin{bmatrix} Y_{1k} \\ Y_{2k} \end{bmatrix} \\ &= \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix} \begin{bmatrix} \Lambda^{-k} Y_{1,0} \\ \Lambda^k Y_{2,0} \end{bmatrix} \end{aligned} \quad (7.94)$$

As k increases, the Λ^{-k} matrix approaches zero (because it is a diagonal matrix whose elements are all less than one in magnitude). Therefore, for large k we obtain

$$\begin{aligned} \begin{bmatrix} Z_k \\ S_k \end{bmatrix} &= \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix} \begin{bmatrix} 0 \\ \Lambda^k Y_{2,0} \end{bmatrix} \\ Z_k &= \Psi_{12} Y_{2k} \\ S_k &= \Psi_{22} Y_{2k} \end{aligned} \quad (7.95)$$

Solving for S_k for large values of k gives

$$S_k = \Psi_{22} \Psi_{12}^{-1} Z_k \quad (7.96)$$

But we also know from Equation (7.82) that

$$S_k = P_k Z_k \quad (7.97)$$

Combining the two previous equations shows that

$$\lim_{k \rightarrow \infty} P_k = \Psi_{22} \Psi_{12}^{-1} \quad (7.98)$$

This gives us a way to determine the steady-state solution of the Riccati equation solution. However, this analysis assumed that Λ was a diagonal matrix with all elements outside the unit circle. In other words, if the Hamiltonian matrix has any eigenvalues with magnitude equal to one, then this analysis falls apart. This gives the following algorithm for computing the steady-state, discrete-time Riccati equation solution.

The Hamiltonian approach to steady-state Kalman filtering

1. Form the Hamiltonian matrix

$$\mathcal{H} = \begin{bmatrix} F^{-T} & F^{-T}H^TR^{-1}H \\ QF^{-T} & F + QF^{-T}H^TR^{-1}H \end{bmatrix} \quad (7.99)$$

For an n -state Kalman filtering problem, the Hamiltonian matrix will be a $2n \times 2n$ matrix.

2. Compute the eigenvalues of $\mathcal{H}$. If any of them are on the unit circle, then we cannot go any further with this procedure; the Riccati equation does not have a steady-state solution.
3. Collect the n eigenvectors that correspond to the n eigenvalues that are outside the unit circle. Put these n eigenvectors in a matrix partitioned as

$$\begin{bmatrix} \Psi_{12} \\ \Psi_{22} \end{bmatrix} \quad (7.100)$$

The first column of this matrix is the first eigenvector, the second column is the second eigenvector, etc. Ψ_{12} and Ψ_{22} are both $n \times n$ matrices.

4. Compute the steady-state Riccati equation solution as

$$P_{\infty}^{-} = \Psi_{22}\Psi_{12}^{-1} \quad (7.101)$$

Note that Ψ_{12} must be invertible for this method to work.

The Hamiltonian approach to steady-state filtering is due to [Vau70], which also derives time-varying DARE solutions using Hamiltonian matrices.

■ EXAMPLE 7.9

Consider the scalar system of Equation (7.36):

$$\begin{aligned} x_{k+1} &= x_k + w_k \\ y_k &= x_k + v_k \\ w_k &\sim N(0, 1) \\ v_k &\sim N(0, 1) \end{aligned} \quad (7.102)$$

We see that $F = H = Q = R = 1$. Substituting these values into the expression for the Hamiltonian matrix gives

$$\begin{aligned} \mathcal{H} &= \begin{bmatrix} F^{-T} & F^{-T}H^TR^{-1}H \\ QF^{-T} & F + QF^{-T}H^TR^{-1}H \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \end{aligned} \quad (7.103)$$